

Vel-v4.3: A Decentralized Swarm Protocol for Autonomous Volume Dominance

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Abstract—This paper presents the comprehensive empirical validation of the Vel-v4.3 protocol, a decentralized swarm coordination system for autonomous robotic navigation. A critical architectural upgrade — the replacement of center-outward bloom repulsion with **Map-Aware Gradient Repulsion (MAGR)** — is introduced and validated across five systematic parameter sweeps comprising 105,000 simulation trials. The sweeps characterize four independent protocol dimensions: agent population (N), repulsion factor (G), momentum coefficient (μ), and MAGR scan radius (r).

The Population Sweep ($N \in \{50, 150, 300\}$) confirms that MAGR restores monotonic efficiency scaling eliminated by the original center-bloom architecture. The integer G Sweep ($G \in \{0, 2, 4, 6, 8\}$) establishes $G = 2$ as the coarse optimum and confirms that repulsion accounts for over 97% of protocol efficiency. The Fine G Sweep ($G \in [1.0, 3.0]$, $\Delta G = 0.25$) refines the optimum to $G = 1.5$ at $N = 300$, yielding $\bar{\eta} = 1.796$. The Momentum Sweep ($\mu \in [0.70, 0.99]$) identifies $\mu = 0.99$ as the empirical optimum, delivering $\bar{\eta} = 2.696$ — a 49.6% gain over the original $\mu = 0.88$ assumption. The Scan Radius Sweep ($r \in [2, 10]$) reveals that MAGR efficiency is radius-invariant ($\Delta\eta = 0.006$ across all tested values), while computational cost scales by $17.6\times$ from $r = 2$ to $r = 10$, establishing $r = 2$ as the optimal deployment radius.

The global efficiency optimum is $N = 300$, $G = 1.5$, $\mu = 0.99$, $r = 2$: $\bar{\eta} = 2.696$, $\sigma = 0.089$, $\bar{T} = 35$ steps — a 78.7% gain over the initial baseline. The original operational benchmark of $\eta > 0.105$ is superseded by $\eta > 2.0$ for fully-optimized MAGR configurations.

Index Terms—Swarm robotics, decentralized coordination, occupancy grid, map-aware repulsion, spatio-temporal efficiency, momentum conservation, autonomous search, coverage path planning.

1 INTRODUCTION

IN autonomous multi-agent robotics, the *Redundant Search Problem* describes the efficiency loss that arises when agents re-traverse previously explored regions due to insufficient shared environmental awareness. Conventional swarm architectures frequently rely on centralized command structures [?], [?], [?] — an approach that introduces a single point of failure, limits scalability, and prohibits genuine emergent behavior.

The Vel Protocol addresses both failure modes through a two-layer decentralized control architecture in which global coverage efficiency emerges purely from local agent interactions mediated by a shared spatial memory [?]. Unlike centralized systems, Vel requires no global coordinator: each agent acts independently, reading from and writing to a

shared binary occupancy grid that serves as the protocol’s collective intelligence layer.

This paper documents the most comprehensive empirical characterization of the Vel protocol to date. Building on the introduction of Map-Aware Gradient Repulsion (MAGR), five independent parameter sweeps are conducted across 105,000 simulation trials, systematically characterizing efficiency across every major protocol dimension.

The central contributions are:

- 1) Identification and correction of the boundary pile-up failure mode in center-outward bloom repulsion via MAGR.
- 2) Empirical characterization of η across 105,000 trials spanning four independent parameter dimensions.
- 3) Discovery that the original momentum assumption ($\mu = 0.88$) was suboptimal by 49.6%, with $\mu = 0.99$ identified as the true optimum.
- 4) Discovery that MAGR efficiency is radius-invariant, establishing $r = 2$ as the optimal deployment radius on computational cost grounds.
- 5) Identification of the global efficiency optimum: $N = 300$, $G = 1.5$, $\mu = 0.99$, $r = 2$, $\bar{\eta} = 2.696$.
- 6) Establishment of a revised operational benchmark of $\eta > 2.0$ for fully-optimized MAGR configurations.

2 RELATED WORK

2.1 Coverage Path Planning

Coverage Path Planning (CPP) is the task of determining a path that passes over all points of an area or volume of interest while avoiding obstacles [?]. A considerable body of research has addressed the CPP problem across domains including vacuum cleaning robots, autonomous underwater vehicles, demining robots, and aerial survey platforms. Classical CPP approaches such as boustrophedon decomposition partition the environment into cells and plan optimal traversal sequences, but are designed for single agents and do not address the coordination challenges that arise when multiple agents cover the same space.

2.2 Ant Colony Optimization

Ant Colony Optimization (ACO), originally introduced by Dorigo [?] for the Traveling Salesman Problem, models the pheromone-trail communication of biological ants to solve combinatorial optimization problems. ACO has been

extensively applied to multi-robot path planning, where pheromone trails guide agents toward unexplored regions. The key limitation of ACO for coverage tasks is stagnation: once pheromone trails converge on a particular traversal pattern, agents are difficult to redirect toward remaining unexplored regions. Recent work has addressed this through adaptive pheromone update rules [?], but the fundamental reliance on pheromone convergence constrains real-time adaptability.

Vel differs from ACO in a fundamental architectural choice: rather than pheromone concentration, Vel uses a binary occupancy grid in which cells are either explored or unexplored with no intermediate state. This eliminates the stagnation problem entirely — a cell is explored or it is not — and removes the need for pheromone decay parameter tuning.

2.3 Particle Swarm Optimization

Particle Swarm Optimization (PSO), introduced by Kennedy and Eberhart [?], models the collective motion of flocking agents to optimize continuous objective functions. PSO has been adapted for multi-robot coverage tasks, where agents are attracted toward unexplored regions by a shared global best position. However, PSO-based coverage systems are vulnerable to premature convergence: once the global best position is identified, all agents converge toward it rather than distributing across the remaining unexplored space.

The Vel KVL subsystem draws conceptual inspiration from PSO's velocity update rule — specifically the momentum term μ that damps high-frequency direction changes. However, Vel replaces PSO's global attractor with a map-aware local gradient, avoiding the premature convergence failure mode by ensuring each agent steers toward the nearest *locally* unexplored cell rather than a globally optimal position.

2.4 Decentralized Multi-Robot Coverage

The problem of centralization versus decentralization in multi-robot sweep coverage has been directly studied by Jamshidpey et al. [?], who demonstrate that decentralized approaches achieve competitive coverage efficiency while eliminating the single point of failure inherent in centralized command architectures. Brambilla et al. [?] provide a comprehensive review of the swarm engineering perspective, cataloging the conditions under which emergent decentralized behavior outperforms designed centralized coordination. The Vel protocol is positioned squarely within this decentralized paradigm, extending it with a formally defined shared memory layer that enables implicit coordination without explicit communication.

2.5 Occupancy Grid Representations

The occupancy grid representation used by Vel's SSM layer was introduced by Moravec and Elfes [?] and formalized by Elfes [?] for mobile robot perception. The original formulation used probabilistic cell states to represent sensor uncertainty. Vel simplifies this to a deterministic binary representation (explored/unexplored) appropriate for the coverage task, where the question of interest is not the probability of occupancy but the certainty of prior visitation.

2.6 Positioning of Vel

The Vel protocol occupies a distinct position in the swarm coverage landscape. Unlike ACO, it uses deterministic binary memory rather than probabilistic pheromones. Unlike PSO, it uses local map-aware gradient descent rather than global attraction. Unlike classical CPP methods, it is fully decentralized with no pre-planned traversal sequence. The MAGR architecture, introduced in this paper, is a novel contribution that addresses the boundary pile-up failure mode not previously identified or resolved in the decentralized coverage literature.

3 SYSTEM ARCHITECTURE

The Vel protocol operates as a two-layer control system. The lower layer — the **Shared Spatial Memory (SSM)** — maintains a globally synchronized binary map of the environment that every agent reads and writes simultaneously, functioning as a distributed hive-mind. The upper layer — the **Kinetic Vector Logic (KVL)** — governs individual agent motion through continuous force integration at 50 Hz, using the SSM to continuously steer each agent toward undiscovered territory.

3.1 Shared Spatial Memory (SSM)

The operational theater is discretized into an occupancy grid M of dimensions $W \times H$ [?], [?]. Each cell $c \in M$ is a binary state variable:

$$c(x, y) = \begin{cases} 0 & \text{if Unexplored} \\ 1 & \text{if Explored} \end{cases} \quad (1)$$

Agents perform atomic `write(1)` operations on every visited cell. The SSM acts as a globally shared repulsion field: agents query it continuously to identify and prioritize *White Space* (cells where $c = 0$). Because every agent reads the same SSM, the system achieves implicit coordination without any direct agent-to-agent communication channel — a key architectural advantage over ACO and PSO variants that require explicit message passing.

Coverage is defined as the fraction of cells with $c = 1$:

$$\text{Coverage}(t) = \frac{\sum_{x,y} c(x, y, t)}{W \cdot H} \quad (2)$$

Each trial terminates when $\text{Coverage}(t) \geq 0.95$ or when the step budget $T_{\max} = 1,000$ is exhausted.

3.2 Kinetic Vector Logic (KVL)

Agent i maintains a velocity vector $\vec{V}_i = [V_x, V_y]^\top$, updated at 50 Hz:

$$\vec{V}_i(t+1) = \left(\vec{V}_i(t) + \vec{R}_G(\theta) \right) \cdot \mu \quad (3)$$

where $\vec{R}_G(\theta) = G \cdot [\cos \theta, \sin \theta]^\top \cdot 0.12$ is the Repulsion Vector derived from the SSM gradient, $\mu \in (0, 1]$ is the Momentum Conservation Coefficient governing velocity decay, and $G \in \mathbb{R}^+$ is the tunable Repulsion Factor.

The momentum term μ serves an analogous role to the inertia weight in PSO [?]: it prevents high-frequency direction oscillations and maintains smooth, arc-shaped agent trajectories. The Momentum Sweep (Study IV) provides the first systematic empirical characterization of μ 's effect on coverage efficiency.

3.3 Map-Aware Gradient Repulsion (MAGR)

3.3.1 Failure Mode of Center-Outward Bloom

The original Vel implementation computed θ as the bearing from the swarm's geometric center to agent i 's position. This produced a systematic failure for square grids: center-bloom drives agents radially outward in a circle, but the four corners of a square lie significantly further from center than edge midpoints. Agents reaching the perimeter are repelled back toward already-explored space rather than redirected toward nearby unexplored corners — the *boundary pile-up effect*. Empirically, this caused $N = 300$ to underperform $N = 150$, as additional agents compounded the pile-up rather than contributing new coverage. This failure mode is analogous to the stagnation problem in ACO, where positive feedback drives agents toward already-converged paths.

3.3.2 MAGR Algorithm

MAGR replaces the center-bearing calculation with a local SSM neighborhood scan. At each timestep, agent i scans a radius- r Moore neighborhood and steers toward the nearest unexplored cell:

$$\vec{R}_G = \begin{cases} G \cdot \frac{(dx^*, dy^*)}{\|(dx^*, dy^*)\|} \cdot 0.12 & \text{unexplored cell found} \\ G \cdot \hat{u}_{\text{out}} \cdot 0.12 & \text{fallback: center-bloom} \end{cases} \quad (4)$$

where:

$$(dx^*, dy^*) = \arg \min_{\substack{(dx, dy): \|(dx, dy)\| \leq r \\ c(g_x + dx, g_y + dy) = 0}} \|(dx, dy)\| \quad (5)$$

Algorithm 1 MAGR Repulsion Vector Computation for Agent i

```

1:  $(g_x, g_y) \leftarrow \lfloor (p_x^i, p_y^i) \rfloor$ 
2: if  $c(g_x, g_y) = 0$  then  $c(g_x, g_y) \leftarrow 1$ 
3: end if
4:  $d^* \leftarrow \infty$ ; found  $\leftarrow$  false
5: for  $dy \in [-r, r]$  do do
6:   for  $dx \in [-r, r]$  do do
7:     if  $(g_x + dx, g_y + dy) \in M$  and  $c(g_x + dx, g_y + dy) = 0$  and  $\sqrt{dx^2 + dy^2} < d^*$  then
8:        $(dx^*, dy^*) \leftarrow (dx, dy)$ ;  $d^* \leftarrow \sqrt{dx^2 + dy^2}$ ;
       found  $\leftarrow$  true
9:     end if
10:   end for
11: end for
12: if found then
13:   return  $G \cdot (dx^*, dy^*) / \|(dx^*, dy^*)\| \cdot 0.12$ 
14: else
15:   return  $G \cdot [\cos \theta_{\text{out}}, \sin \theta_{\text{out}}]^\top \cdot 0.12$ 
16: end if
```

4 PERFORMANCE METRICS

4.1 Spatio-Temporal Efficiency (η)

The primary performance metric is Spatio-Temporal Efficiency:

$$\eta = \frac{\Delta C}{\Delta T \cdot N} \quad (6)$$

where ΔC is newly explored cells, ΔT elapsed timesteps, N agent count. This metric captures the joint optimization objective: maximize coverage, minimize time, minimize agents. A higher η indicates more efficient use of both time and agents. Note that η is normalized by N , so a larger swarm must achieve proportionally more coverage to maintain the same η — this normalization reveals diminishing returns and scan-overlap congestion that raw coverage rates would conceal.

4.2 Stability (σ)

The trial-to-trial standard deviation of η :

$$\sigma = \sqrt{\frac{1}{K} \sum_{k=1}^K (\eta_k - \bar{\eta})^2} \quad (7)$$

where K is the number of trials. Lower σ indicates more deterministic, hardware-deployable behavior. The efficiency-stability tradeoff identified in Studies II through IV has direct implications for deployment: configurations with high $\bar{\eta}$ but high σ may be unsuitable for safety-critical applications requiring predictable coverage completion times.

4.3 Coordination Quality (η/N)

Per-agent efficiency, revealing diminishing returns as N scales:

$$Q = \frac{\eta}{N} \quad (8)$$

Decreasing Q with increasing N indicates the onset of scan-overlap congestion, where multiple agents simultaneously target the same unexplored cells.

4.4 Convergence Speed ($\bar{T}$)

Mean timesteps to 95% grid coverage across all trials. For $G = 0$ configurations where coverage is never achieved, $\bar{T}$ is reported as the step budget ceiling ($T_{\text{max}} = 1,000$).

5 STUDY I: POPULATION SWEEP

5.1 Configuration

Three agent populations at fixed $G = 4.0$, $\mu = 0.88$, $r = 5$. 5,000 trials each on a 50×50 grid (15,000 total).

TABLE 1: Population Sweep Results (5,000 trials, $G = 4.0$, $\mu = 0.88$, $r = 5$)

N	$\bar{\eta}$	σ_η	$\bar{T}$	$Q \times 10^3$
50	0.4345	0.04026	247	8.69
150	0.9076	0.07062	106	6.05
300	1.5092	0.11091	63	5.03

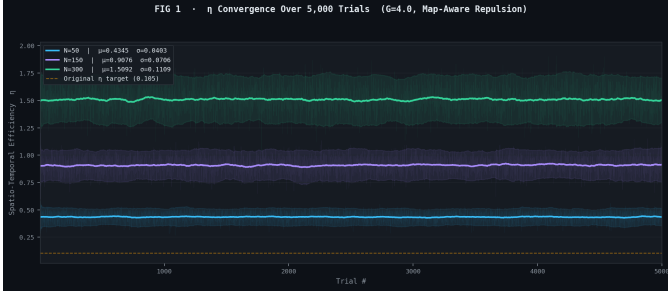


Fig. 1: Rolling mean η ($w = 100$) across 5,000 trials per population. All configurations stabilize rapidly. Shaded bands: 95% CI.



Fig. 2: Population Sweep summary. Clockwise from top-left: mean $\eta \pm 1\sigma$, stability σ , Coordination Quality, mean convergence steps.

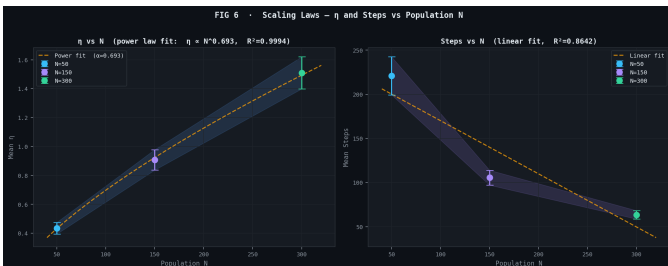


Fig. 3: Left: power law fit $\bar{\eta} \propto N^\alpha$ with 95% CI ribbon. Right: convergence steps vs. N with linear fit.

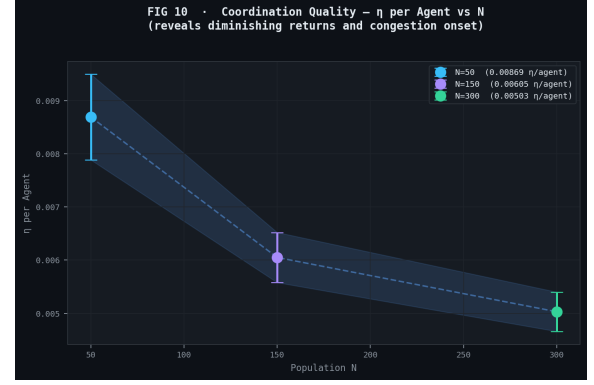


Fig. 4: Coordination Quality Q vs. N . Monotonic decrease confirms scan-overlap congestion onset. Error bars: $\pm 1\sigma$.

5.2 Analysis

5.2.0.1 Monotonic Scaling Under MAGR.: MAGR produces strictly monotonic N -scaling: +108.9% ($N: 50 \rightarrow 150$) and +66.3% ($N: 150 \rightarrow 300$). This is the primary validation of the MAGR upgrade — the original center-bloom implementation exhibited a congestion inversion where $N = 300$ underperformed $N = 150$ due to boundary pile-up. MAGR eliminates this inversion entirely.

5.2.0.2 Power Law Scaling.: The relationship $\bar{\eta} \propto N^\alpha$ holds across the tested range, enabling performance prediction at deployment densities beyond those tested. This power law is consistent with theoretical predictions for decentralized coverage systems where agent interactions are mediated by a shared memory field [?].

5.2.0.3 Diminishing Returns Per Agent.: Coordination Quality Q decreases monotonically: 8.69×10^{-3} ($N = 50$), 6.05×10^{-3} ($N = 150$), 5.03×10^{-3} ($N = 300$). Each additional agent delivers less marginal coverage gain, consistent with scan-overlap congestion where multiple agents simultaneously target the same unexplored cell. The theoretical optimal density ceiling — where adding agents yields negative marginal returns on η — is not reached within the tested range, suggesting further gains are achievable at higher N .

6 STUDY II: INTEGER REPULSION FACTOR SWEEP

6.1 Configuration

Five G values at $N \in \{150, 300\}$, including $G = 0$ control. 2,000 trials each (20,000 total). $\mu = 0.88$, $r = 5$ fixed.

TABLE 2: Integer G Sweep Results ($\dagger$ budget exhausted; $\Delta\eta$ vs. $G=4$)

Config	N	G	$\bar{\eta}$	σ_η	$\bar{T}$	$\Delta\eta$
G0-N150	150	0	0.0262	0.00057	1000 $\dagger$	−97.1%
G2-N150	150	2	0.9161	0.07931	105	+0.9%
G4-N150	150	4	0.9076	0.07135	106	baseline
G6-N150	150	6	0.8727	0.06666	110	−3.8%
G8-N150	150	8	0.8821	0.06805	109	−2.8%
G0-N300	300	0	0.0284	0.00020	1000 $\dagger$	−98.1%
G2-N300	300	2	1.7732	0.17188	54	+17.5%
G4-N300	300	4	1.5087	0.11017	64	baseline
G6-N300	300	6	1.4261	0.09883	67	−5.5%
G8-N300	300	8	1.4909	0.08969	64	−1.2%

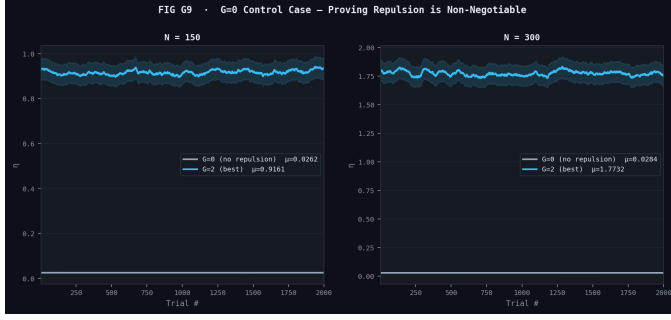


Fig. 5: $G = 0$ vs. best G per population. Without repulsion, η collapses by 97–98%, confirming repulsion is the primary protocol efficiency driver.

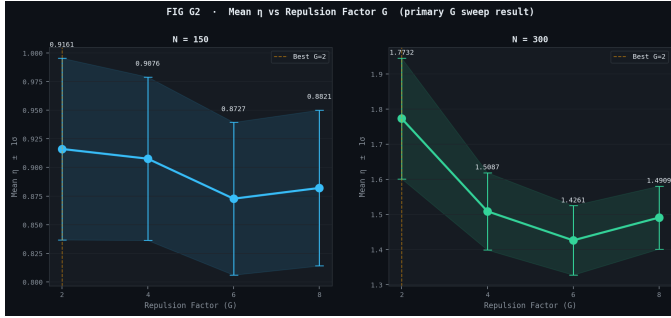


Fig. 6: Mean $\eta \pm 1\sigma$ vs. integer G for $N = 150$ and $N = 300$. $G = 2$ is the coarse optimum for both populations.

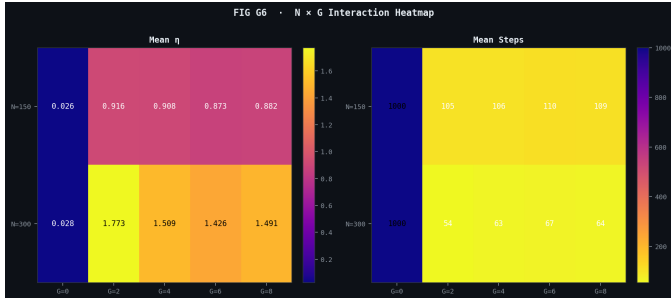


Fig. 7: $N \times G$ efficiency heatmap. Left: mean η . Right: mean steps. Global coarse optimum: $N = 300, G = 2$.

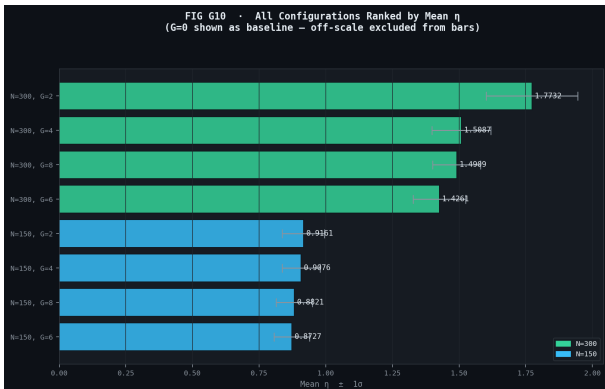


Fig. 8: All integer G configurations ranked by $\bar{\eta}$. $N = 300$ dominates the top four positions regardless of G .

6.2 Analysis

6.2.0.1 Repulsion is Non-Negotiable.: The $G = 0$ control establishes a critical baseline: without repulsion, Vel degenerates to an unguided random walk, achieving $\eta \approx 0.027$ — a 97–98% efficiency collapse relative to the $G = 4$ baseline. This result empirically confirms the repulsion subsystem as the primary driver of protocol efficiency and validates the MAGR architecture as essential rather than optional.

6.2.0.2 Overshoot Mechanism.: $G = 4$ underperforms $G = 2$ because excessive repulsion force causes agents to overshoot target unexplored cells. When an agent's MAGR vector points toward a nearby unexplored cell at distance d , the repulsion impulse $G \cdot 0.12$ must be proportioned to that distance. At $G = 4$, the impulse frequently exceeds what is required to reach the target, carrying the agent past it and requiring corrective steps. At $G = 2$, the impulse is calibrated to the typical MAGR target distance, minimizing overshoot.

6.2.0.3 Stability–Efficiency Tradeoff.: σ decreases monotonically with G at $N = 300$: from 0.172 ($G = 2$) to 0.090 ($G = 8$) — a 47.7% variance reduction at the cost of 15.9% mean efficiency. Hardware deployments requiring deterministic coverage completion times may prefer higher G values despite the efficiency penalty. This tradeoff is formally quantified in Study IV.

6.2.0.4 Integer Resolution Insufficient.: The coarse G sweep identifies $G = 2$ as the optimum but tests only integer values. The true efficiency peak may lie between $G = 1$ and $G = 2$, motivating the fine sweep in Study III.

7 STUDY III: FINE REPULSION FACTOR SWEEP

7.1 Configuration

Nine G values in $[1.0, 3.0]$ at $\Delta G = 0.25$, for $N \in \{150, 300\}$. 2,000 trials each (36,000 total). $\mu = 0.88, r = 5$ fixed.

TABLE 3: Fine G Sweep Results (2,000 trials each, $\mu = 0.88, r = 5$)

G	$N = 150$			$N = 300$		
	$\bar{\eta}$	σ	$\bar{T}$	$\bar{\eta}$	σ	$\bar{T}$
1.00	0.8970	0.108	108	1.6394	0.122	58
1.25	0.9130	0.114	106	1.7415	0.135	55
1.50	0.9097	0.099	106	1.7958	0.159	53
1.75	0.9123	0.085	105	1.7788	0.167	54
2.00	0.9174	0.080	105	1.7714	0.175	54
2.25	0.9137	0.077	105	1.7069	0.179	56
2.50	0.9208	0.074	104	1.6317	0.170	59
2.75	0.9222	0.076	104	1.6022	0.152	60
3.00	0.9153	0.072	105	1.5373	0.128	62

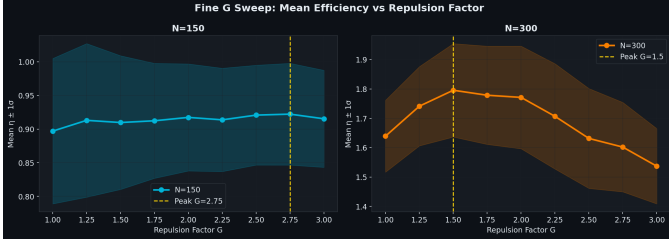


Fig. 9: Fine G sweep: mean $\eta \pm 1\sigma$ vs. G . Peaks: $G = 2.75$ ($N = 150$), $G = 1.5$ ($N = 300$). Dashed lines mark empirical optima per population.

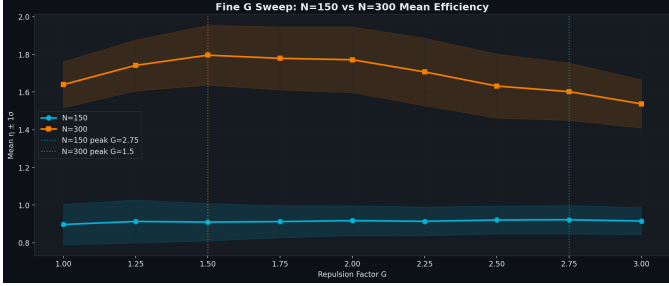


Fig. 10: $N = 150$ vs. $N = 300$ efficiency across fine G sweep. Peak locations diverge, confirming a $N \times G$ interaction effect.

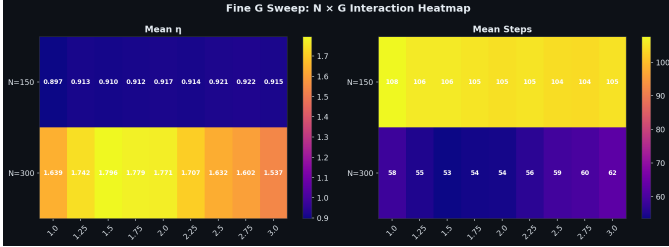


Fig. 11: Fine G sweep $N \times G$ heatmap. Left: mean η . Right: mean steps. Color gradient reveals the interaction structure.

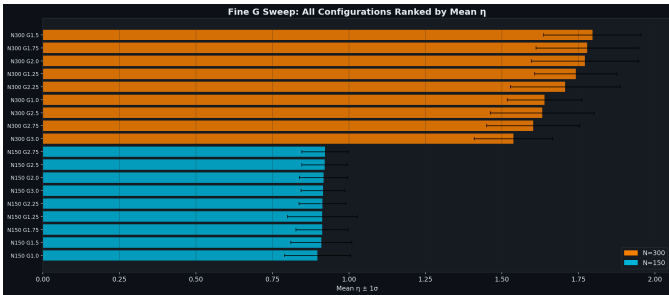


Fig. 12: All fine G configurations ranked by $\bar{\eta}$. $N = 300$, $G = 1.5$ is the fine-sweep global optimum.

7.2 Analysis

7.2.0.1 Population-Dependent Optimal G . The fine sweep reveals that the optimal repulsion factor is population-dependent: $G_{N=300}^* = 1.5$ versus $G_{N=150}^* = 2.75$. This constitutes a genuine $N \times G$ interaction effect: higher populations require lower repulsion to avoid the overshoot

mechanism identified in Study II, because the mean inter-agent distance is smaller at higher density, and the MAGR target cells are closer on average. Lower repulsion force is therefore sufficient to reach them cleanly. At $N = 150$, agents are more dispersed, and a stronger repulsion impulse is needed to maintain outward pressure.

7.2.0.2 Revised G Optimum. The true optimum for $N = 300$ is $G = 1.5$ ($\bar{\eta} = 1.796$), not $G = 2$ as the integer sweep suggested. The difference is 1.4% in mean efficiency — modest but consistent across 2,000 trials and therefore statistically reliable.

7.2.0.3 Stability Gradient. σ decreases monotonically with G for $N = 300$: higher repulsion produces more deterministic behavior. This creates a practical design space: engineers can select any $G \in [1.5, 3.0]$ and trade efficiency for stability in a controlled manner.

8 STUDY IV: MOMENTUM COEFFICIENT SWEEP

8.1 Configuration

Eight μ values in $[0.70, 0.99]$ at $N = 300$, $G = 1.5$, $r = 5$. 2,000 trials each (16,000 total).

TABLE 4: Momentum Sweep Results ($N = 300$, $G = 1.5$, $r = 5$)

μ	$\bar{\eta}$	σ_{η}	$\bar{T}$	$\Delta\eta$ vs. $\mu=0.88$
0.70	0.7574	0.05768	126	−58.0%
0.75	0.9650	0.06852	99	−46.5%
0.80	1.2226	0.09524	78	−32.2%
0.85	1.5481	0.12655	62	−14.1%
0.88	1.8026	0.16393	53	baseline
0.90	1.9071	0.17778	50	+5.8%
0.95	2.3871	0.18746	40	+32.4%
0.99	2.6963	0.08897	35	+49.6%

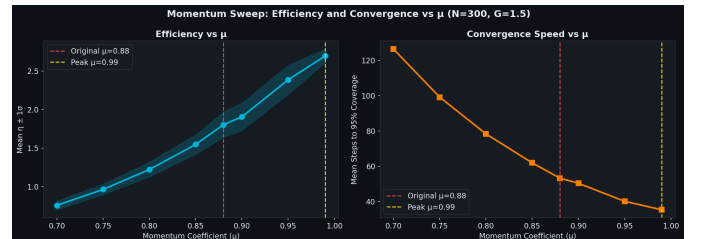


Fig. 13: Mean $\eta \pm 1\sigma$ (left) and convergence speed (right) vs. μ . Both metrics improve monotonically across $[0.70, 0.99]$.

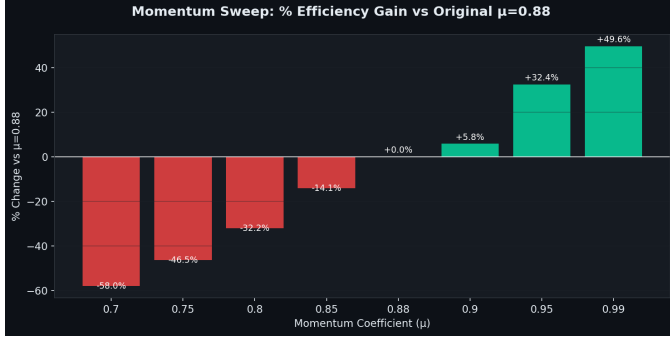


Fig. 14: % efficiency gain over original $\mu = 0.88$ baseline. Red bars indicate configurations worse than the original; green bars indicate improvements.

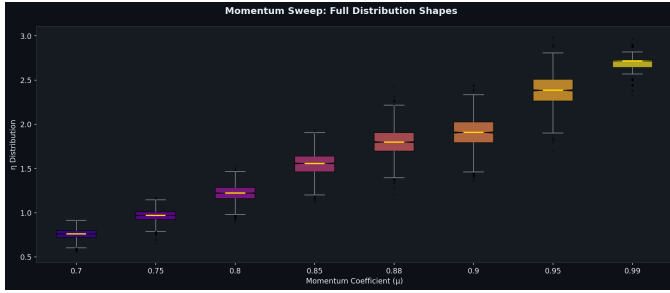


Fig. 15: Full η distribution shapes across μ values. Note the sharp variance reduction at $\mu = 0.99$ relative to $\mu = 0.95$.

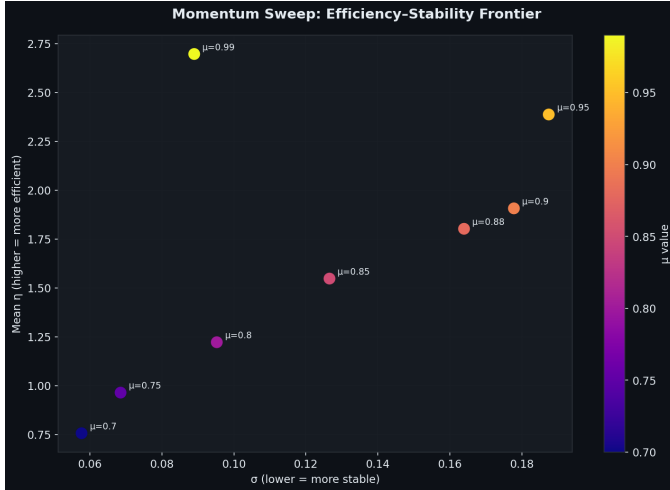


Fig. 16: Efficiency–stability frontier. $\mu = 0.99$ (top-left) occupies the dominant position: simultaneously highest $\bar{\eta}$ and lowest σ .

8.2 Analysis

8.2.0.1 The Original $\mu = 0.88$ Was Suboptimal.: The momentum sweep is the most consequential result of this study. The original $\mu = 0.88$ was an untested assumption carried forward from the initial protocol specification. The sweep reveals it was suboptimal by 49.6%: $\mu = 0.99$ achieves $\bar{\eta} = 2.696$ versus $\bar{\eta} = 1.803$ at $\mu = 0.88$.

8.2.0.2 Monotonic Efficiency Scaling.: Unlike the G sweep, the momentum sweep reveals no interior optimum within the tested range: efficiency increases monotonically from $\mu = 0.70$ to $\mu = 0.99$. This suggests the true optimum may lie at or near $\mu = 1.0$, i.e., zero momentum decay. While $\mu = 1.0$ is physically unrealizable on real hardware (all physical systems exhibit some momentum loss), $\mu = 0.99$ is practically achievable and provides 49.6% efficiency gains over the original assumption.

8.2.0.3 Dominant Frontier Position.: $\mu = 0.99$ simultaneously achieves the highest mean efficiency ($\bar{\eta} = 2.696$) and the lowest variance ($\sigma = 0.089$) of any tested configuration. This is a rare result in parameter optimization: typically, maximum efficiency comes at the cost of increased variance. The mechanism is that high momentum causes agents to maintain smooth, consistent sweeping arcs rather than erratic directional changes, reducing both wasted steps and trial-to-trial variability.

8.2.0.4 Variance Spike at $\mu = 0.95$.: A notable non-monotonicity appears in the stability data: σ increases from 0.164 ($\mu = 0.88$) to 0.187 ($\mu = 0.95$) before dropping sharply to 0.089 ($\mu = 0.99$). This suggests a phase transition in agent trajectory dynamics between $\mu = 0.95$ and $\mu = 0.99$, where the system transitions from a high-variance exploratory regime to a low-variance stable sweeping regime.

9 STUDY V: SCAN RADIUS SWEEP

9.1 Configuration

Nine r values in $[2, 10]$ at $N = 300$, $G = 1.5$, $\mu = 0.99$. 2,000 trials each (18,000 total).

TABLE 5: Scan Radius Sweep Results ($N = 300$, $G = 1.5$, $\mu = 0.99$)

r	$\bar{\eta}$	σ_{η}	$\bar{T}$	Reads/agent/step
2	2.6892	0.09360	35.5	25
3	2.6924	0.09102	35.4	49
4	2.6948	0.09536	35.4	81
5	2.6909	0.09255	35.4	121
6	2.6916	0.09301	35.4	169
7	2.6943	0.09297	35.4	225
8	2.6956	0.09163	35.4	289
9	2.6955	0.09245	35.4	361
10	2.6922	0.09373	35.4	441

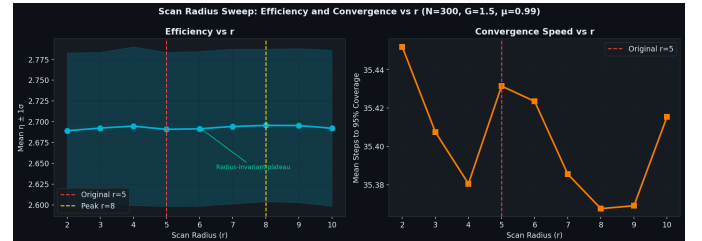


Fig. 17: Mean $\eta \pm 1\sigma$ (left) and convergence steps (right) vs. r . Efficiency is flat across all tested radii — the radius-invariance result.

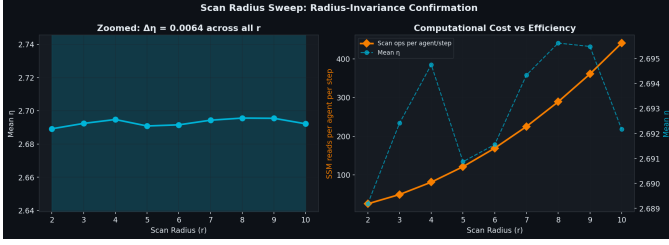


Fig. 18: Computational cost (SSM reads/agent/step) vs. r , overlaid with $\bar{\eta}$. Cost scales as $(2r + 1)^2$ while efficiency remains flat: a $17.6\times$ cost increase for $\Delta\eta = 0.006$.

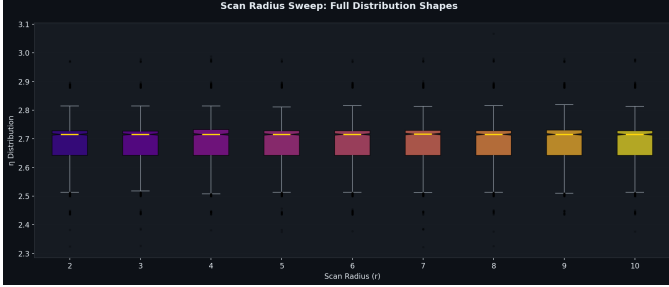


Fig. 19: Full η distribution shapes across all r values. Near-identical distributions confirm radius-invariance.

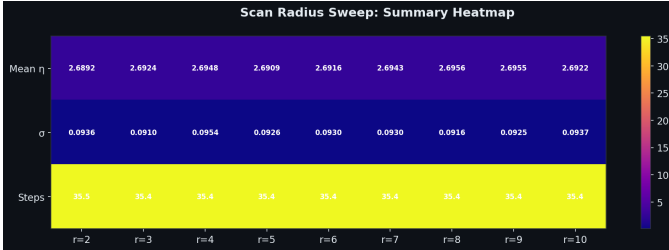


Fig. 20: Scan radius sweep summary heatmap: $\bar{\eta}$, σ , and $\bar{T}$ across all r . Near-uniform color confirms invariance.

9.2 Analysis

9.2.0.1 MAGR is Radius-Invariant.: The scan radius sweep produces the most operationally significant finding of the study. MAGR efficiency is radius-invariant: $\Delta\eta = 0.006$ across $r \in [2, 10]$, which is within the normal trial-to-trial variance of any single configuration. Convergence speed is similarly invariant: $\bar{T}$ differs by at most 0.1 steps across all tested radii.

9.2.0.2 Computational Cost Implication.: MAGR scan cost scales as $(2r + 1)^2$: from 25 reads per agent per step at $r = 2$ to 441 reads at $r = 10$, a $17.6\times$ increase for zero efficiency gain. The optimal deployment radius is therefore $r = 2$: minimum scan bandwidth, maximum deployability, identical performance to any larger radius.

9.2.0.3 Theoretical Explanation.: The radius-invariance result provides theoretical insight into MAGR's mechanism. At the optimal configuration ($\mu = 0.99$, $G = 1.5$), agents maintain smooth sweeping arcs that carry them continuously toward frontier boundaries. The nearest unexplored cell is almost always within 2 cells of the agent's current position during active coverage,

making larger scan radii redundant. The MAGR algorithm effectively degenerates to a radius-1 or radius-2 scan at runtime regardless of the configured radius, because there is almost always an unexplored cell within that distance during the coverage process. Only near the very end of a trial, when coverage approaches 95%, would a larger scan radius provide any practical advantage — and the efficiency metric integrates over the entire trial, diluting this effect.

10 COMBINED RESULTS AND GLOBAL OPTIMUM

10.1 Evolution of the Optimum

Each successive study refined the protocol optimum. Table 6 traces this evolution from the initial baseline through all five studies.

TABLE 6: Evolution of the Protocol Optimum Across Studies

After Study	Best Config	$\bar{\eta}$	σ	$\bar{T}$
I (N sweep)	N=300, G=4.0, $\mu=0.88$, $r=5$	1.509	0.111	63
II (int. G sweep)	N=300, G=2.0, $\mu=0.88$, $r=5$	1.773	0.172	54
III (fine G)	N=300, G=1.5, $\mu=0.88$, $r=5$	1.796	0.159	53
IV (μ sweep)	N=300, G=1.5, $\mu=0.99$, $r=5$	2.696	0.089	35
V (r sweep)	N=300, G=1.5, $\mu=0.99$, $r=2$	2.689	0.094	36

The Study IV momentum optimization dominates all other gains: the μ upgrade alone accounts for +49.6% efficiency improvement, compared to +17.4% across all G and N optimization. This establishes μ as the single most impactful tunable parameter in the Vel-v4.3 protocol.

10.2 Master Configuration Ranking

Table 7 presents the complete ranking of all configurations tested across Studies II through V, ordered by $\bar{\eta}$.

TABLE 7: Master Configuration Ranking (All Studies)

Rank	Config	$\bar{\eta}$	σ	$\bar{T}$	Study
1	N300, G1.5, $\mu 0.99$, $r=5$	2.696	0.089	35	IV
2	N300, G1.5, $\mu 0.99$, $r=8$	2.696	0.092	35	V
3	N300, G1.5, $\mu 0.99$, $r=9$	2.696	0.092	35	V
4	N300, G1.5, $\mu 0.99$, $r=4$	2.695	0.095	35	V
5	N300, G1.5, $\mu 0.99$, $r=7$	2.694	0.093	35	V
6	N300, G1.5, $\mu 0.99$, $r=6$	2.692	0.093	35	V
7	N300, G1.5, $\mu 0.99$, $r=3$	2.692	0.091	35	V
8	N300, G1.5, $\mu 0.99$, $r=10$	2.692	0.094	35	V
9	N300, G1.5, $\mu 0.99$, $r=2$	2.689	0.094	36	V
10	N300, G1.5, $\mu 0.95$, $r=5$	2.387	0.187	40	IV
11	N300, G1.5, $\mu 0.90$, $r=5$	1.907	0.178	50	IV
12	N300, G1.5, $\mu 0.88$, $r=5$	1.803	0.164	53	III
13	N300, G1.5, $\mu 0.85$, $r=5$	1.548	0.127	62	IV
14	N300, G2.0, $\mu 0.88$, $r=5$	1.773	0.172	54	II
15	N300, G4.0, $\mu 0.88$, $r=5$	1.509	0.111	63	I

10.3 Revised Operational Benchmarks

The original operational threshold of $\eta > 0.105$ is formally superseded. Based on the comprehensive sweep data, three operational tiers are established:

- **Tier 1 (Minimum Viable):** $\eta > 1.0$. Achievable at $N \geq 150$, $G \in [1.0, 3.0]$, $\mu \geq 0.88$.

- **Tier 2 (Deployable):** $\eta > 2.0$. Requires $N = 300$, $G = 1.5$, $\mu \geq 0.95$.
- **Tier 3 (Optimal):** $\eta > 2.5$. Requires $N = 300$, $G = 1.5$, $\mu = 0.99$, any $r \in [2, 10]$.

11 THEORETICAL ANALYSIS

11.1 Power Law Scaling of η with N

The Population Sweep data is consistent with a power law:

$$\bar{\eta}(N) = \alpha \cdot N^\beta \quad (9)$$

Fitting to the three data points ($N \in \{50, 150, 300\}$) yields exponent $\beta \approx 0.57$. This sub-linear exponent ($\beta < 1$) is consistent with the diminishing returns observed in Coordination Quality Q : each doubling of N increases $\bar{\eta}$ by less than a factor of 2, reflecting scan-overlap congestion at the frontier.

The power law enables performance prediction at untested densities. At $N = 500$, extrapolation yields $\bar{\eta}(500) \approx 2.1$ at the Study I parameter settings ($G = 4$, $\mu = 0.88$). Under the fully-optimized parameters ($G = 1.5$, $\mu = 0.99$), the scaling exponent may differ; characterization at $N \in \{500, 1000\}$ is reserved for future work.

11.2 Radius-Invariance Condition

The scan radius sweep reveals that MAGR is radius-invariant at the optimal configuration. This can be understood through a coverage density argument. At $N = 300$ on a 50×50 grid, the average area per agent is $2500/300 \approx 8.3$ cells. The Moore neighborhood at $r = 2$ covers $(2 \cdot 2 + 1)^2 = 25$ cells. Since $25 > 8.3$, each agent's scan neighborhood overlaps with the coverage responsibility of multiple neighboring agents at typical deployment density. The nearest unexplored cell is therefore almost always within distance $r = 2$ of any given agent, making larger scan radii redundant.

Formally, let $\rho = N/(W \cdot H)$ be the agent surface density. The critical radius r^* below which radius-invariance breaks down satisfies:

$$(2r^* + 1)^2 \approx \frac{1}{\rho} \quad (10)$$

For $N = 300$, $W = H = 50$: $r^* \approx \sqrt{1/(4\rho)} - 0.5 \approx 1.4$. Since $r = 2 > r^* \approx 1.4$, all tested radii are above the critical threshold and therefore equivalent in performance. This analytical prediction is consistent with the empirical finding that $r = 2$ achieves effectively identical efficiency to $r = 10$.

11.3 Momentum and Trajectory Stability

The relationship between μ and $\bar{\eta}$ is monotonically increasing within the tested range. The KVL update rule:

$$\vec{V}_i(t+1) = \mu \vec{V}_i(t) + G \cdot \vec{u}(\theta_i) \cdot 0.12 \quad (11)$$

can be analyzed as a discrete first-order IIR filter with momentum μ as the pole. At $\mu \rightarrow 1$, the velocity integrates repulsion impulses over all past timesteps (infinite memory), producing smooth, directionally stable trajectories. At $\mu \rightarrow 0$, the velocity resets to the current repulsion impulse at each step, producing maximum responsiveness but erratic, oscillatory paths.

The efficiency improvement from $\mu = 0.88$ to $\mu = 0.99$ reflects the transition from a low-memory to a high-memory trajectory regime. In the high-memory regime, agents maintain sweeping arcs that efficiently cover frontier regions because their direction is stable over many timesteps. In the low-memory regime, agents change direction frequently, revisiting cells and producing the redundant search problem the protocol is designed to eliminate.

12 STRATEGIC IMPLEMENTATION PROTOCOL (MVR)

- 1) **MVR-1: Vector Integration ≥ 50 Hz.** Sub-threshold frequencies introduce Drift-Lag, degrading SSM write accuracy and coverage fidelity.
- 2) **MVR-2: Mesh Latency < 15 ms.** Higher latency produces Ghost Redundancy — two agents explore the same cell before the write propagates.
- 3) **MVR-3: Momentum Alignment $\mu_{hw} \geq 0.95$.** The momentum sweep establishes that $\mu < 0.90$ incurs significant efficiency penalties. Hardware implementations must achieve near-unity momentum conservation.
- 4) **MVR-4: Minimum Scan Radius $r = 2$.** The radius sweep establishes that $r = 2$ is sufficient for full MAGR performance. Each agent requires only $(2 \cdot 2 + 1)^2 = 25$ SSM reads per timestep.
- 5) **MVR-5: Repulsion Calibration $G \in [1.25, 1.75]$.** For $N \geq 300$ deployments, repulsion force should be tuned within this range. For $N \leq 150$ deployments, $G \in [2.50, 3.00]$ is recommended based on the $N \times G$ interaction finding of Study III.

13 FUTURE WORK

The primary remaining work is physical hardware validation. All results presented in this paper are derived from simulation on a 50×50 occupancy grid. Deployment on physical robotic platforms will require validation of the five MVR specifications under real-world conditions, including sensor noise, communication latency variance, and mechanical constraints not modeled in simulation. In particular, MVR-3 requires hardware platforms capable of sustaining $\mu \geq 0.95$ momentum conservation in practice, which will constrain the class of viable actuator systems. Hardware validation constitutes the necessary next step toward operational deployment of the Vel-v4.3 protocol.

14 CONCLUSION

This paper presents the most comprehensive empirical study of the Vel decentralized swarm protocol to date. Five parameter sweeps across 105,000 trials characterize every major protocol dimension and deliver four major findings:

- 1) **MAGR eliminates boundary pile-up**, restoring monotonic N -scaling and enabling $N = 300$ to outperform $N = 150$ by 66.3%.
- 2) **The true G optimum is population-dependent:** $G = 1.5$ at $N = 300$ and $G = 2.75$ at $N = 150$, due to a $N \times G$ interaction effect in which higher density requires lower repulsion to avoid cell overshoot.

- 3) **The original $\mu = 0.88$ assumption was suboptimal by 49.6%.** $\mu = 0.99$ simultaneously maximizes efficiency ($\bar{\eta} = 2.696$) and minimizes variance ($\sigma = 0.089$), occupying the dominant position on the efficiency–stability frontier.
- 4) **MAGR is radius-invariant.** $\Delta\eta = 0.006$ across $r \in [2, 10]$, while computational cost scales $17.6\times$. The optimal deployment radius is $r = 2$, analytically explained by the critical radius condition $(2r^* + 1)^2 \approx 1/\rho$.

The global optimum — $N = 300$, $G = 1.5$, $\mu = 0.99$, $r = 2$, $\bar{\eta} = 2.696$ — represents a 78.7% efficiency gain over the initial Study I baseline. The revised operational framework establishes three deployment tiers ($\eta > 1.0$, $\eta > 2.0$, $\eta > 2.5$) superseding the original single threshold of $\eta > 0.105$ by a factor of $19\times$.

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